AGH University of Krakow

Documentation for the project

Weather station

from the course

DESIGN LABORATORY

Electronics and Telecommunications (PL), 3rd year

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1. Project description

Project involves the design and implementation of a measurement and data logging system based on ATmega328pb microcontroller. The system is intended for measuring temperature and pressure using dedicated digital sensors: DS18B20 and BMP280. The captured data can be displayed on a 7-segment display and transmitted to a PC via USB-UART interface.

The system is designed in a modular way and implemented in stages, which allows for its gradual evolution and easy diagnostics of individual functional blocks. The project includes both hardware parts, such as schematic design and PCB layout, and software parts responsible for sensor handling, data presentation, data storage and user communication.

2. User manual

After a successful compilation, we can see the temperature from the BMP280 sensor displayed on a 7-segment display connected to the microcontroller. After a short press and hold of the S1 button on the keyboard, the pressure value is being displayed. Repeating this action, we get a temperature value from DS18B20 sensor, then we can come back to the first read. The UART-USB converter is used in order to enable communication between two PC`s by UART standard. We can receive the data from the sensors in terminal and additionally send messages from PC1 to PC2 (echo action).

3. Source files

Microchip Studio project contains:

- main.c, main.h - Main program of the project
- diodes.c, diodes.h - Led diodes control
- display.c, display.h - 7-segment display code
- uart.c, uart.h - Communication by UART standard
- sensors.c, sensors.h - BMP280, DS18B20 temperature and pressure sensors control

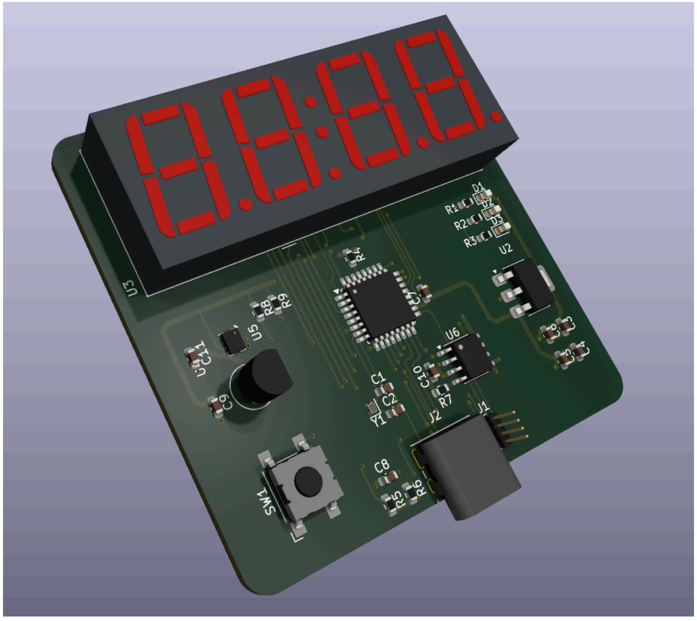
4. User Interface

The custom user interface designed for the weather station was written in the Python programming language. The application enables communication with the measurement device via serial COM port, while the current connection status is clearly presented to the user through a *connected* / *disconnected* status message.

The interface displays real-time measurement data captured from the weather station sensors, including temperature and pressure. These values are presented in standard physical units. Additionally, the application includes a log section that displays raw data frames received from the microcontroller. The interface also allows for sending custom text commands, enabling full diagnostics of the UART communication.

The current interface architecture provides a solid basis for further system development, such as adding data logging to files or visualizing measurements using charts.

5. PCB



The full hardware documentation is located on github.

https://github.com/adbik4/DL-FinalProject